



(b) A user-defined record structure is used to store the nodes of the linked list in part (a).

Complete the diagram, using your answer for part (a).

RootPtr	Index	LeftPtr	Data	RightPtr
0	0		Red	
	1		Green	
	2		Yellow	
	3		Blue	
	4		Orange	
	5		Indigo	
FreePtr	6		Violet	
	7			

[4]

(c) The linked list in part (a) is implemented using a 1D array of records. Each record contains a left pointer, data and a right pointer.

The following pseudocode represents a function that searches for an element in the array of records `BinTree`. It returns the index of the record if the element is found, or it returns a null pointer if the element is **not** found.

Complete the pseudocode for the function.

```

FUNCTION SearchTree(Item : STRING) .....

    NowPtr ← .....

    WHILE NowPtr <> -1

        IF ..... THEN

            NowPtr ← BinTree[NowPtr].LeftPtr

        ELSE

            IF BinTree[NowPtr].Data < Item THEN

                .....

            ELSE

                RETURN NowPtr

            ENDIF

        ENDIF

    ENDWHILE

    RETURN NowPtr

ENDFUNCTION
    
```

[4]

